FLYRANK FRONTEND AI ENGINEERING TRACK · FE-AA1 BUILD+

Buttons with a Brain: Motion & State Micro-interactions

Intentional state choreography across 6 distinct states, compositor-only transform and opacity transitions, interruptible lifecycle handling, and strict reduced-motion accessibility.

CANDIDATE	ASSIGNMENT CODE	PRIMARY AI PLATFORM	STACK & TECH
KM Siam	**FE-AA1 (Week 6 · 2h)**	**Google Antigravity**	**React 19 · Next.js 15 · Tailwind**

Executive Certification: Engineered the production-ready `SmartButton` component (`src/components/ui/SmartButton.jsx`) demonstrating complete micro-interaction state choreography across 6 states: *idle, hover/focus, loading, success, error, disabled*. Validated with an interactive workbench at `/micro-interactions` containing both a Primary Chat sender and a Secondary Generator button. 63/63 automated tests passed.

1. Six-State Micro-Interaction Choreography

The component avoids abrupt swaps by guaranteeing that every state change is a mathematically continuous transition:

State	Visual Treatment	Motion & Property Transition	Duration & Easing
1. Idle	Brand color, crisp icon + label, subtle shadow	Baseline position, `scale: 1.0`, `opacity: 1.0`	Default resting state
2. Hover / Focus	White/10 highlight sheen, visible focus ring	Subtle scale micro-lift, focus ring offset 2px	`150ms ease-out`
3. Active / Click	Tactile depth depression	`scale(0.98)` on mouse down	`100ms ease-in`
4. Loading	Spinning loader replaces icon, label updates	Label translates, spinner smoothly fades in (opacity 0 → 1)	`200ms ease-out`
5. Success	Emerald green badge, checkmark icon	Checkmark scales to `1.1`; auto-reverts to idle in 1.8s	`200ms ease-out`
6. Error	Rose red badge, retry icon, physics shake	Horizontal keyframe shake (translateX ±4px), persists for retry	`350ms cubic-bezier`

2. Duration & Easing Rationale (Why these choices?)

1. Fast 200ms Window

Human perceptual reaction time averages 150–250ms. An interaction transition of **200ms** feels snappy and immediate to power users without jarring the visual cortex.

2. ease-out Landing

Standard transitions use `ease-out` so the change initiates with urgency and decelerates smoothly into place, creating physical authenticity.

3. Physics-Based Shake

The error state uses `cubic-bezier(0.36, 0.07, 0.19, 0.97)` over 350ms to simulate the physical damping of an elastic spring when rejecting an input.

3. System Flexibility: Two Unified Buttons (/micro-interactions)

The system proves it is a repeatable design language by powering two distinct components on the live demo page:

Button A: Primary Chat Co-Pilot ("Send Message")

Powers message transmission in the AI chat studio. Transitions to "Synthesizing...", morphs to "Sent to Stream", or triggers "Retry Message" upon network drop.

Button B: Secondary Generative Tool ("Run Audit")

Powers prompt engineering evaluation in Generative UI. Shares identical timing curves, reduced motion rules, and ARIA live regions.

4. Compositor Performance & Reduced Motion Guarantees

- **Zero Layout Thrash:** Transitions animate exclusively `transform` and `opacity`. No height, width, margin, or layout recomputations occur (CLS = 0.00).
- **Interruptible Lifecycle:** Spam-clicking during in-flight transitions is safely locked via debounced state guards without dropping event handlers or leaking timeouts.
- **prefers-reduced-motion:** Under `motion-reduce`, all transforms and keyframe shakes are completely eliminated via `motion-reduce:transform-none` and `motion-reduce:animate-none`, while retaining vital color and text feedback.
- **Keyboard & Screen Readers:** Visible `focus-visible:ring-2` focus ring and `aria-live="polite"` ensure instant accessibility for keyboard and screen reader users.

5. Automated Verification & Deployment

Test Suite	Tests Passed	Coverage Focus
`tests/smart-button.test.js`	**6 / 6 Passed**	State invariants (idle, loading, success, error, disabled), compositor safety, reduced motion, ARIA regions, demo page.
Total Automated Suite (npm test)	**63 / 63 Passed**	Zero regressions across all repository milestones.

Verification Link & Live URL:

- Live Demo URL: https://flyrank-internship-front-end-ai-engineering.vercel.app/micro-interactions
- Source File: `src/components/ui/SmartButton.jsx`
- Next.js Build: 15/15 static and dynamic pages compiled in 4.1s with 0 errors.

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